

Exam 1 — Residual Networks: Gradient Flow & Identity Mappings

The mathematics of why depth becomes trainable

Theme. We build the residual block from first principles and study the single phenomenon that makes very deep networks optimizable: the additive identity path. From scratch we derive the forward telescoping identity, the backward gradient-flow formula, the spectral behaviour of a residual block's Jacobian, the parameter/compute accounting of basic vs. bottleneck blocks, and the “unrevealed” ensemble interpretation. We explicitly defer to later exams: the internals of convolution and batch normalization, optimizer dynamics (SGD/Adam), data augmentation, normalization-free training in full generality, and any attention/transformer material. Here, $\mathcal{F}$ is treated as an abstract differentiable residual map except where convolution arithmetic is required.

Duration. 3 hours. **Total.** 110 pts.

Rules. Any result you prove may be reused later. State every assumption. Define notation before using it. Partial rigor earns partial credit; hand-waving where rigor is expected does not. A correct final formula with no derivation earns at most a third of the marks.

Notation & standing assumptions (read before starting)

- A network is a composition of L *residual blocks* indexed $l = 0, 1, \dots, L - 1$, with input x_0 and final representation x_L .
 - Block l computes $x_{l+1} = x_l + \mathcal{F}(x_l; W_l)$, where the *residual function* $\mathcal{F}(\cdot; W_l) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ has parameters W_l and is differentiable in both arguments. By default $\dim x_l = d$ for all l .
 - A *plain* (non-residual) block is $x_{l+1} = \mathcal{F}(x_l; W_l)$.
 - $\mathcal{L} \in \mathbb{R}$ is a scalar loss computed from x_L (through a fixed head). Upstream gradients are row vectors; $\frac{\partial \mathcal{L}}{\partial x} \in \mathbb{R}^{1 \times d}$.
 - For $g : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $\frac{\partial g}{\partial x} \in \mathbb{R}^{d \times d}$ is the Jacobian. I is the $d \times d$ identity, $\|\cdot\|$ the spectral norm, and $\sigma_{\max}(\cdot)$ the largest singular value.
 - A reference residual block: the input x_l splits into (i) the residual branch $\mathcal{F}(\cdot; W_l)$ and (ii) an untouched identity branch; the two are summed to give x_{l+1} .
 - Convolution accounting (Parts A, D): a conv with C_{in} input channels, C_{out} output channels, kernel $k \times k$, output resolution $H \times W$, stride 1, no bias, has $C_{\text{in}}C_{\text{out}}k^2$ parameters and $HWC_{\text{in}}C_{\text{out}}k^2$ multiply-accumulates (MACs).
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Part A — The degradation problem & the block (approx. 25 min, 16 pts)

A1. (Degradation puzzle) — 4 pts

He et al. (2015) observed that a plain 56-layer CNN had *higher training error* than a plain

20-layer CNN on CIFAR-10.

- (a) Explain precisely why this observation cannot be attributed to overfitting.
- (b) A colleague argues: “the 56-layer net contains the 20-layer net as a special case, so its training error should be at least as low.” State the theoretical construction that makes this argument valid, and explain why optimization can still fail to realize it.

A2. (Residual reformulation) — 4 pts

Let the ideal map a block should implement be $H(x)$, and write $\mathcal{F}(x) = H(x) - x$.

- (a) Express H via $\mathcal{F}$. What must $\mathcal{F}$ output if H is exactly the identity?
- (b) Argue qualitatively why driving stacked weight layers toward the zero map is an easier optimization target than driving them to reproduce the identity.

A3. (Counting) — 5 pts

A basic block uses two 3×3 convolutions, each mapping C channels to C channels at resolution $H \times W$. Ignore biases and normalization parameters.

- (a) Give the learnable parameter count of the block.
- (b) Give the MAC count of one forward pass on a single image.
- (c) State how each quantity scales in C and in $H \times W$, and which dominates memory vs. compute.

A4. (Typing the shortcut) — 3 pts

When $\mathcal{F}$ changes channel count or spatial resolution, the identity shortcut $x_{l+1} = x_l + \mathcal{F}(x_l)$ is ill-typed.

- (a) Name the two standard remedies.
- (b) For each, state whether it introduces learnable parameters, and give one sentence on the trade-off between them.

Part B — Gradient flow through residual connections (*approx. 55 min, 30 pts*)

B1. (Scalar Jacobian) — 4 pts

Consider depth- L scalar linear blocks $x_{l+1} = x_l + w_l x_l = (1 + w_l)x_l$, with $x_0 \in \mathbb{R}$ and $\mathcal{L} = \frac{1}{2}(x_L - t)^2$.

- (a) Show that $\frac{\partial x_L}{\partial x_0} = \prod_{l=0}^{L-1} (1 + w_l)$.
- (b) Show the plain analogue $x_{l+1} = w_l x_l$ gives $\frac{\partial x_L}{\partial x_0} = \prod_{l=0}^{L-1} w_l$.

B2. (Vanishing at initialization) — 6 pts

Suppose the w_l are i.i.d. with $\mathbb{E}[w_l] = 0$ and $\text{Var}(w_l) = \epsilon^2$ for small ϵ .

- (a) Approximate $\log \left| \frac{\partial x_L}{\partial x_0} \right|$ for the residual network to first non-trivial order in ϵ , and describe the typical gradient magnitude at x_0 as L grows.
- (b) Do the same for the plain network and contrast. Which network suffers vanishing gradients at initialization, and why?
- (c) State the qualitative conclusion this toy model is meant to convey about real residual networks.

B3. (Weight gradient) — 4 pts

For the scalar residual network of B1, derive $\frac{\partial \mathcal{L}}{\partial w_l}$ in closed form. Identify the multiplicative factor through which the upstream error $\partial \mathcal{L} / \partial x_L$ reaches w_l , and explain in one sentence why it is well-behaved near initialization.

B4. (Telescoping) — 4 pts

Return to the general block $x_{l+1} = x_l + \mathcal{F}(x_l; W_l)$. Prove by telescoping that for any $l < L$,

$$x_L = x_l + \sum_{i=l}^{L-1} \mathcal{F}(x_i; W_i).$$

B5. (Identity-mapping gradient formula) — 6 pts

- (a) Using the chain rule on B4, prove

$$\frac{\partial \mathcal{L}}{\partial x_l} = \frac{\partial \mathcal{L}}{\partial x_L} \left(I + \frac{\partial}{\partial x_l} \sum_{i=l}^{L-1} \mathcal{F}(x_i; W_i) \right).$$

- (b) Interpret the role of the I term for gradient propagation.
- (c) Under what (pathological) condition could $\partial \mathcal{L} / \partial x_l$ still vanish, and why is that condition unlikely to hold across an entire mini-batch?

B6. (Plain product bound) — 3 pts

For the plain network, $\frac{\partial x_L}{\partial x_l} = \prod_{i=l}^{L-1} \frac{\partial \mathcal{F}_i}{\partial x_i}$. Suppose $\sigma_{\max}(\partial \mathcal{F}_i / \partial x_i) \leq s$ for all i .

- (a) Bound $\left\| \frac{\partial x_L}{\partial x_l} \right\|$.
- (b) State the qualitative regimes for $s < 1$, $s = 1$, $s > 1$, and explain why the residual network does not obey such a clean s^L law.

B7. (Spectral view of one block) — 3 pts

Write $\frac{\partial x_{l+1}}{\partial x_l} = I + J_l$ with $J_l = \partial \mathcal{F}_l / \partial x_l$, and assume $\|J_l\| \leq \rho < 1$.

- (a) Show every singular value of $I + J_l$ lies in $[1 - \rho, 1 + \rho]$.
- (b) Explain how this simultaneously prevents per-block collapse and unbounded blow-up of the Jacobian.

Part C — Identity mappings & pre-activation (*approx. 25 min, 15 pts*)

C1. (Post-activation breaks telescoping) — 4 pts

The original block applies a nonlinearity after the sum: $x_{l+1} = \phi(x_l + \mathcal{F}(x_l))$ with $\phi = \text{ReLU}$. Redo the telescoping of B4 for this block, show the clean additive form $x_L = x_l + \sum \mathcal{F}$ no longer holds, and explain via B5 what is lost when ϕ sits on the main path.

C2. (Pre-activation clean path) — 4 pts

The pre-activation block (He et al. 2016) is $x_{l+1} = x_l + \mathcal{F}(x_l)$ where $\mathcal{F}$ begins with BN and ReLU. Argue that the map from x_0 to x_L is an exact identity term plus residual terms with no nonlinearity gating the skip path, and state the consequence for $\partial\mathcal{L}/\partial x_l$.

C3. (Sign preservation) — 3 pts

A ReLU on the main path can output only non-negative values, whereas a clean identity path carries signed information across many blocks. Give a concrete reason why preserving sign across depth matters for representation learning.

C4. (Debugging scenario) — 4 pts

A practitioner reports that a 200+ layer post-activation ResNet trains noticeably worse than its pre-activation twin of identical width and parameter count.

- (a) Using C1–C3, explain the most likely mechanism.
- (b) Name one quantity you would log during training to confirm the diagnosis, and the signature you would expect to see.

Part D — Bottleneck blocks & computational complexity (*approx. 30 min, 17 pts*)

A *bottleneck* block (ResNet-50/101/152) has input/output width $4C$ and is the composition: 1×1 conv ($4C \rightarrow C$), then 3×3 conv ($C \rightarrow C$), then 1×1 conv ($C \rightarrow 4C$), all at resolution $H \times W$, stride 1, no bias.

D1. (Bottleneck parameters) — 4 pts

Compute the parameter count of the bottleneck block as a multiple of C^2 by summing the three convolutions.

D2. (MAC accounting) — 5 pts

- (a) Compute the bottleneck block's total MACs as a multiple of $HW C^2$.
- (b) Compute the MACs of a basic two- 3×3 block of equal I/O width $4C$, and show it equals $288 HW C^2$.

D3. (The ratio) — 3 pts

Form the ratio (basic $\div$ bottleneck) as a single number. Interpret: at fixed representational width $4C$, how much compute does the bottleneck save, and where does the saving come from?

D4. (Design logic) — 3 pts

Explain the rationale of “squeeze with 1×1 , mix spatially in the cheap low-dimensional space with 3×3 , then expand with 1×1 .” What does this trade away relative to a basic block, and when might that trade-off hurt?

D5. (Shortcut accounting) — 2 pts

In a stage that changes width or stride, the projection shortcut is a single 1×1 conv ($4C_{\text{in}} \rightarrow 4C_{\text{out}}$). Argue whether its cost is asymptotically negligible compared with the main path, and why.

Part E — ResNets as ensembles: the unraveled view (*approx. 25 min, 15 pts*)

Consider n residual blocks $x_i = x_{i-1} + f_i(x_{i-1})$, $i = 1, \dots, n$, with input x_0 .

E1. (Unrolling) — 4 pts

Expand x_2 , then x_3 , fully (substitute the recursion and distribute). Write each as an explicit sum of terms, where each term corresponds to choosing, at every block, either the identity branch or the residual branch.

E2. (Counting paths) — 4 pts

- (a) Prove the fully unraveled x_n is a sum of 2^n terms (“paths”).
- (b) Show the number of paths passing through exactly k residual functions is $\binom{n}{k}$.

E3. (Path-length distribution) — 3 pts

Weighting all paths equally makes the path length k distribute as $\text{Binomial}(n, \frac{1}{2})$. Compute the mean path length, and describe qualitatively how concentrated the distribution is around it for large n .

E4. (Interpretation & caveat) — 4 pts

- (a) Veit et al. (2016) found that deleting one residual block at test time barely changes accuracy, while deleting a layer of a plain VGG-style net is catastrophic. Explain this through the path picture.
- (b) State one reason the “ensemble of 2^n independent networks” reading is an approximation rather than a literal truth.

Part F — Open-ended / research-style (*approx. 25 min, 17 pts*)

F1. (Variance growth $\Rightarrow$ normalization is unavoidable) — 6 pts

Model block l as adding a residual whose output is, at initialization, approximately zero-mean and independent of x_l with per-coordinate variance σ_l^2 . Assume approximate independence.

- (a) Show $\text{Var}(x_L^{(j)}) \approx \text{Var}(x_0^{(j)}) + \sum_{l=0}^{L-1} \sigma_l^2$, i.e. activation variance grows roughly linearly in depth.

- (b) Explain why this growth endangers a deep stack, and how (i) Batch/Layer Norm and (ii) zero-initializing the last BN scale γ of each residual branch each tame it. For (ii), what is the effective network at step 0 of training, and why is that a desirable starting point?

F2. (Continuous-depth limit) — 6 pts

Insert a step size: $x_{l+1} = x_l + h \mathcal{F}(x_l; W_l)$.

- (a) Identify the ODE of which this is the forward (explicit) Euler discretization, and the meaning of h , L , and total time $T = Lh$.
- (b) Predict what happens to a trained network's accuracy at test time if you (i) double the number of blocks by interpolating weights, or (ii) halve h . State the assumptions under which your prediction holds and where it may fail.
- (c) Give one engineering advantage and one disadvantage of replacing the discrete stack with a black-box ODE solver (memory, latency, adaptivity, debuggability).

F3. (Design challenge) — 5 pts

You must train a 1000-layer image classifier on a fixed compute budget. Propose an architecture and training recipe grounded in this exam's results. Justify your choices on at least three of: shortcut type, block type (basic vs. bottleneck), placement of nonlinearities, residual-branch initialization, normalization, learning-rate warmup — naming the failure mode each choice guards against. Briefly state how you would monitor training to detect those failure modes early.

End of Exam 1. Attempt as much as you can before requesting corrections. The grading scheme, common-mistake notes, and the geometric / probabilistic intuition behind each step will be supplied — but only after you have tried.